

of timing resistors connected between pins 15 and 16 of HT12E and HT12D, for the given supply voltages, can be determined from the graphs given in the datasheet of the respective chips. The resistor values used in the circuit here are chosen for approximately 3kHz frequency for encoder HT12E and 150 kHz for decoder HT12D at a V_{DD} of 5V.

Decoder HT12D receives data from HT12E on its D_N pin serially. If the transmitted address matches the address of the decoder four times in succession, valid transmission pin (V_N) becomes high. The data from pins AD8 through AD11 of the HT12E appears on pins D8 through D11 of the HT12D.

Transmitter unit

Fig. 2 shows the transmitter circuit with SM TX-433 RF module (TX1). TX1 is an AM/ASK transmitter module operating at 433 MHz. AT89C2051 is a low-voltage, high-performance CMOS 8-bit microcontroller. It has 2 kB of Flash, 128 bytes of RAM, 15 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, a precision analogue comparator, on-chip oscillator and clock circuitry.

Power-on reset is provided by the combination of resistor R3 and capacitor C1. Switch S17 is used for manual reset. A 12MHz crystal along with two 33pF capacitors provides the basic clock frequency for operation.

The receiver address to be transmitted can be set with the help of 8-way DIP switch S18. Port pins P1.0 through P1.7 of the microcontroller are interfaced with the keypad. Pins P3.0 and P3.2 through P3.5 are interfaced with TE pin and data inputs AD8 through AD11 of encoder HT12E.

When all switches (S1 through S16) are opened on the keypad, the microcontroller pulls the TE pin as well as data input pins AD8 through AD11 to logic 1. If any switch is closed, the microcontroller pulls the corresponding data pin along with TE pin to logic 0. When switch S1 is closed, the microcontroller makes pin 10 (AD8) and pin 14 (TE) of encoder HT12E low, and

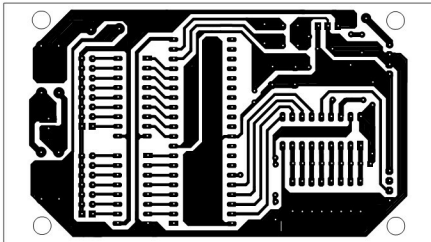


Fig. 6: An actual-size, single-side PCB for the receiver circuit (Fig. 3)

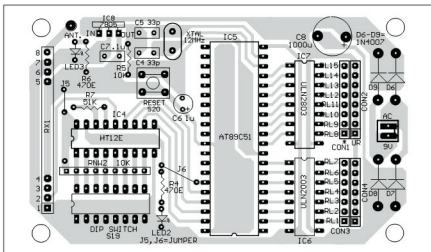


Fig. 7: Component layout for the PCB in Fig. 6

logic 0 is transmitted through TX1. The other data pins of encoder HT12E will be in logic 1 state in this case. LED1 glows to indicate transmission enabled.

Receiver unit

Fig. 3 shows the receiver circuit with SM RX-433 RF module. AT89C51 is a low-power, high-performance CMOS 8-bit microcontroller. It has 4 kB of Flash, 128 bytes of RAM, 32 I/O lines, two 16-bit timers/counters, five-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

Power-on reset is provided by the combination of resistor R5 and capacitor C6. Switch S20 is used for manual reset. A 12MHz crystal along with two 33pF capacitors provides the basic clock frequency to microcontroller AT89C51.

Address lines of the encoder (IC1) and the decoder (IC4) should be identical for data reception in the receiver. Here addresses are made identical through switches S18 and S19. When any of the keys on the keypad is closed, the corresponding data pin of the decoder goes low. When any data is received, valid transmission pin (V_N) goes high as indicated by LED2.

Data outputs D8 through D11 of HT12D are connected to port pins P0.0 through P0.3 of the microcontroller. The microcontroller receives the decoded data and controls the corresponding relay through relay drivers ULN2003 and ULN2803. The device to be controlled is connected to the relay contacts. Unregulated power supply is used for relays.